



AJITH SAMPATH

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CORE EXPERTISE

- **Scientific programming** in Python for large-scale data analysis
- **Radio astronomy** and 21cm cosmology (calibration and signal extraction)
- **Machine learning** for scientific data (classification, signal detection, foreground removal).
- Development of **scalable pipelines and algorithms** for large datasets (HIRAX analysis pipeline)

TECHNICAL SKILLS

- **Programming:** Python (NumPy, SciPy, Pandas, Matplotlib)
- **Machine Learning:** Scikit-learn, PyTorch; classification, signal detection, feature extraction, foreground removal (21cm cosmology)
- **Data Processing:** Large-scale data analysis, scalable pipeline development, data cleaning, and transformation
- **Scientific Computing:** Numerical methods, statistical analysis, signal processing
- **High-Performance Computing:** Parallel computing (MPI), HPC clusters, job scheduling (SLURM)
- **Radio Astronomy Tools:** CASA (calibration, imaging, simulation), WSCLEAN (wide-field imaging), AOFlagger (RFI detection and mitigation), CST Studio Suite (electromagnetic RF simulations)
- **Tools:** Git, Linux, LaTeX

PROFESSIONAL EXPERIENCE

POSTDOCTORAL FELLOW

Jun 2025 – Present

Department of Theoretical Physics, University of Geneva, Switzerland

- SNSF Lead Agency grant (No. 10001432)
- Developed an empirical primary beam model for high-precision HIRAX calibration
- Contributed to large-scale simulations and data analysis pipelines in HIRAX
- Applied machine learning for foreground modeling and mitigation in 21cm cosmology

INTERN

Jan 2020 – Jun 2020

IIT Indore, India

- Performed imaging-based 21cm power spectrum estimation using MWA data

VISITING STUDENT

Jul 2019 – Dec 2019

Indian Institute of Astrophysics, India

- Developed a CASA-based imaging pipeline for the GRAPH telescope
- Built a GUI analysis tool for the GLOSS spectrometer

EDUCATION

UNIVERSITY OF KWAZULU-NATAL, SOUTH AFRICA | *PhD* 2021 – 2024

- Thesis title: Beam Calibration and Foreground Leakage in 21cm Intensity Mapping - Application to HIRAX.
- Supervisor: Prof. Kavilan Moodley
- NRF - TWAS Doctoral Fellowship

UNIVERSITY OF MADRAS, CHENNAI, INDIA | | *MSc Theoretical Physics* 2017 – 2019

- Related Coursework: Astrophysics, Theoretical Cosmology, Numerical Methods.
- Final semester short Project: HI observations of Barred Spiral Galaxies.

BHARATHIDASAN UNIVERSITY, TAMIL NADU, INDIA | | *BSc Physics* 2014 – 2017

- Gold Medalist in Physics - A.V.C. College (Autonomous).

AWARDS

- **"Biggest Impact"** award for Oral Presentation - with cash prize ZAR 5000 2023
Postgraduate Research and Innovation Symposium 2023,
University of KwaZulu-Natal, South Africa.
- **"Winner"** of BRICS Astronomy Hackathon on Machine Learning 2023
BRICS Astronomy meeting, Capetown, South Africa.

LANGUAGE PROFICIENCY

- **English** (Academic IELTS 7.5 - **C1**), French (A1-Learning) and Tamil (Native language).
- Had a **German** introductory course in my Master's.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

IMPRS-HD "AI for Astronomy" Summer School, Heidelberg, Germany. Sep 2025
BRICS Astronomy Hackathon, IDIA, Cape Town, South Africa. Oct 2023
CASA-VLBI Workshop - JIVE, Virtual. Nov 2020
Synthesis Imaging Workshop - NRAO, Virtual. Jul 2020
Observing First Billion years of the Universe School, IIT Indore, India Jan 2020
Solar Astronomy Winter School, Kodaikanal Radio Observatory, India. Jan 2019
"Astrophysics: An observational view of the Universe" Summer Fellowship,
Indian Academy of Sciences, Bengaluru, India. May-Jun 2018

PUBLICATIONS

1. Primary Beam Chromaticity in HIRAX: I. Characterization from Simulations and Power Spectrum Implications 2024
Ajith Sampath, Devin Crichton, Kavilan Moodley, et al.
The Astrophysical Journal - DOI: 10.3847/1538-4357/ae1b89
2. The HIRAX: Overview and Status Update. 2024
Anthony Walters, ... **Ajith Sampath** ... et al. 18th European Conference on Antennas and Propagation
(EuCAP), Glasgow, UK, 2024, pp. 1-5, DOI:10.23919/EuCAP60739.2024.10501639

3. HIRAX: 256-element array status and overview. 2022
*Devin Crichton, ... **Ajith Sampath** ... et al. Journal of Astronomical Telescopes, Instruments, and Systems*
 Vol. 8, Issue 1, 011019 <https://doi.org/10.1117/1.JATIS.8.1.011019>
4. Antenna characterization for the HIRAX experiment. 2022
*Emily R.Kuhn, ... **Ajith Sampath** ... et al. Ground-based and Airborne Telescopes IX. Vol. 12182.*
 SPIE, 2022. <https://doi.org/10.1117/12.2627856>

KEY PROJECTS

- **HIRAX Primary Beam characterization:** Analytically modeled, EM-simulated, and measured antenna patterns, and characterized the impact on the power spectrum space.
- **HIRAX Data Analysis Pipeline:** Contributed to the development of scalable data analysis pipelines for large radio astronomy datasets. Implemented algorithms for signal extraction and foreground mitigation in 21cm cosmology
- **Machine Learning for Foreground Mitigation:** Developed and applied ML models for classification and removal of foreground contamination. Improved accuracy of cosmological signal recovery from noisy data.
- **GRAPH Imaging Pipeline (CASA):** Built an automated imaging pipeline for the GRAPH solar radio telescope using CASA. Processed and analyzed solar interferometric data.
- **GLOSS Spectrometer Analysis Tool:** Developed a GUI-based tool for data visualization and analysis of spectrometer observations - including solar bursts model fitting.

CONFERENCE CONTRIBUTIONS

1. The significance of characterizing the sidelobe systematics in 21cm Intensity Mapping experiments: The HIRAX case. 2024
Ajith Sampath, Devin Crichton, & Kavilan Moodley. DOI: 2024IAUGA..32P.517S
2. Full-wave analysis of mutual coupling in the HIRAX radio telescope. 2022
*Q Gueuning, D Crichton, **A Sampath**, AK Brown, E de Lera Acedo & K Moodley.*
International Conference on Electromagnetics in Advanced Applications (ICEAA), Cape Town, South Africa, 2022, pp. 288-288, DOI: 10.1109/ICEAA49419.2022.9899905.

SELECTED CONTRIBUTED TALKS

- **Swiss Cosmology Days** 2026 — Precision Cosmology meets Imperfect Instruments
- **Cosmology on Alps** 2026 — HIRAX Primary Beam: Challenges and Updates
- **SKACH Winter Meeting** 2026 — HIRAX Overview and Current Status
- **Swiss SKA Days** 2025 — Primary Beam Chromaticity in HIRAX
- **SARAO Postgraduate Conference** 2023&2024 — Beam Systematics and Primary Beam Modelling
- **UKZN PRIS Symposium** 2023 — Instrument Characterization (Biggest Impact Award)
- **Cosmology on Safari.** 2023 - An Analytical approach to characterize Beam Systematics in HIRAX

INVITED TALKS

- **Invited Talk - Alumni Lecture series** Jan 2019
Invited by the department of Physics, A.V.C. College for a lecture on "Introduction to Physics of Cosmos".

MENTORSHIP & COMMUNITY ENGAGEMENT

- **Student supervision** 2026
Supervised a Master's student on a Semester project titled "Cosmology with Fast Radio Bursts" at the University of Geneva
Student: Swan Dubief (Swan.Dubief@etu.unige.ch)
- **Organised Department Seminars** 2023-2024
Co-organizer of Astronomy Research Center Weekly seminars for three semesters at the University of KwaZulu-Natal, South Africa.
- **Medical Data Analysis** 2023-2024
Worked on ICU Heart Rate and Blood pressure data analysis on the KREATE community engagement program with eThekweni Hospital, Durban, South Africa.
- **Led the Weekly KREATE Computer Literacy workshop** 2023
For underprivileged high school students from Lindelani Youth Forum and Community Life Center, Durban, South Africa.
- **Co-led two KREATE Machine Learning workshops** 2023
For students and industry staff hosted by SmartXchange, Durban, South Africa.

PERSONAL INTERESTS

Chess Active player on Chess.com with an interest in strategic problem-solving.

Boxing & Fitness Amateur Boxing training (Boxing Club Genevois, Geneva).

Photography Travel photography enthusiast with a focus on digital post-processing.

REFEREES

- **Prof. Martin Kunz**
University of Geneva, Geneva, Switzerland.
✉ Martin.Kunz@unige.ch
- **Prof. Kavilan Moodley**
University of KwaZulu-Natal, Durban, South Africa.
✉ Moodleyk41@ukzn.ac.za